

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 12 · LECTURE NOTES

Learning in Games

What adaptive agents actually reach.

MODULE OUTCOMES

- Explain no-regret learning and its guarantee.
- State what no-regret dynamics converge to.
- Explain why that answers Module 03's objection.
- Explain best-response dynamics and when they converge.
- Simulate learning agents against a mechanism.

1 No-regret learning

AN ALGORITHM HAS NO REGRET IF ITS AVERAGE PAYOFF APPROACHES THAT OF THE BEST FIXED ACTION IN HINDSIGHT

Regret is the gap between the payoff you actually received and the payoff the single best fixed action would have received — chosen with full hindsight over the whole sequence of rounds, which is a demanding benchmark and is nonetheless achievable.

And no-regret means that gap, divided by the number of rounds, goes to zero as the horizon grows — which is achieved by multiplicative weights and its many relatives (CSCE 658 Module 11's randomised weighted majority, CSCE 669 Module 09's online convex optimisation — the same algorithm appearing for the third time in this program).

Crucially it requires no assumption whatsoever about the environment — the other players may be adversarial, adaptive, irrational, or not players at all, and the guarantee still holds, which is what makes it usable as a behavioural model.

Which makes it a far more defensible behavioural assumption than equilibrium: it says only that agents are not persistently leaving money on the table — which is modest, plausible, and consistent with agents who do not know they are in a game at all.

2 Where it converges

- If every player uses a no-regret algorithm, the empirical distribution of joint play converges to the set of coarse correlated equilibria — not to Nash — a weaker and larger set, and one that is computable by linear programming, unlike Nash (Module 03).

- **With a stronger guarantee — no swap regret, which is also achievable — the play converges to the set of correlated equilibria, which is a tighter set, still weaker than Nash, still computable, and still reached by simple adaptive agents.**
- **And in two-player zero-sum games the time-averaged play approaches the minimax value — which recovers the Nash prediction in exactly the case where Nash is computable (Module 03 §4), and that coincidence is not a coincidence.**
- **Simple adaptive agents reach correlated equilibrium, which is computable — while Nash, which they do not reach, is not. The computable concept is the one that is actually reached, which is a satisfying piece of structure and a good argument for taking computational constraints seriously as modelling constraints.**

WHICH ANSWERS MODULE 03'S OBJECTION RATHER THAN EVADING IT

Module 03 said that computing a Nash equilibrium is intractable, so predicting that agents play one is unjustified — and this module says that agents reach a different solution concept, which is tractable, and reach it by simple and well-understood means.

So the prediction is rescued by weakening it — which is the right move, and is considerably more honest than assuming agents solve a PPAD-complete problem between rounds.

And Module 04 §3's smoothness bounds apply directly to the no-regret outcomes — so the price of anarchy guarantees survive the weakening — which is the result that ties this course's analysis half together and is why the smoothness framework was such a significant development.

Which means the analysis half of this course rests on learning rather than on equilibrium — a reorientation worth stating plainly, and one the field made deliberately in response to the complexity results rather than ignoring them.

3 Best response dynamics

- Each agent in turn switches to its best response against the current profile of everybody else — which is the obvious thing to simulate and is what Project 1 uses to find an equilibrium of a congestion game.
- It converges in potential games (Module 05 §2) — because every improving move strictly decreases the potential, which is bounded — and that is the main positive case, covering routing, load balancing, and congestion generally.
- And it can cycle forever in general games — rock-paper-scissors is the two-line example, with each agent endlessly best-responding to the last move — and nothing in the dynamics prevents it.
- Plus the order of play matters: simultaneous updates can oscillate where sequential updates converge, because simultaneous best responses can overshoot together — an implementation detail with real consequences for a simulation's conclusions.
- So the practical rule: use best-response dynamics where a potential function exists, and no-regret otherwise — which is clean, and is what Project 2's simulation should follow.

4 Using it

SIMULATE LEARNING AGENTS AGAINST YOUR MECHANISM, BECAUSE HONEST AGENTS WILL NOT TELL YOU MUCH

A mechanism tested with agents who report truthfully tests nothing — truthfulness was precisely the property you were trying to establish, so assuming it begs the question, and a simulation built that way will always succeed.

So run no-regret or best-response learners against it, and see what they find — which is Project 2's requirement and is the cheapest attack available on your own design, requiring no cleverness about where the vulnerability might be.

And compare the resulting outcome to the honest-agent case: the gap between them is what strategic behaviour costs your design, measured rather than assumed, and it is the number to report.

Which is this course's version of CSCE 628 Module 13's negative control — test the system under conditions where it might fail, and report what happened, whatever it was — and it is the same argument: a test that cannot fail is not evidence.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Nisan et al., chapter 4 (free PDF)	§§1 and 2 — learning, regret, and the convergence results, with the proofs.
Roughgarden — Twenty Lectures, lectures 17 to 18 (free)	§2 — no-regret dynamics and the connection to the price of anarchy bounds, which is the key link.
Blum & Mansour — From external to internal regret (free)	§2's swap-regret result , which is what gets you to correlated equilibrium.
Cesa-Bianchi & Lugosi — Prediction, Learning, and Games	§1 — the regret framework in full, and the same material CSCE 669 Module 09 uses. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Define regret** and compute it for a short action sequence.
- 2 **Implement multiplicative weights** and verify the regret bound empirically.
- 3 **Run it against an adversarial opponent** and confirm the guarantee holds.
- 4 **Run two no-regret learners against each other** in a small game and plot the empirical distribution.
- 5 **Check whether it is a correlated equilibrium.**
- 6 **Run them in a zero-sum game** and watch the average approach minimax.
- 7 **Implement best-response dynamics** in a potential game and plot the potential.
- 8 **Run it on rock-paper-scissors** and observe the cycle.
- 9 **Compare simultaneous and sequential updates** on the same game.
- 10 **Attack one of your own mechanisms** with a learner and report the gap.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define regret and no-regret.
- 2 What does the guarantee assume about the environment?
- 3 Why is it a better behavioural assumption than equilibrium?
- 4 What do no-regret dynamics converge to?
- 5 What does swap-regret buy, and what about zero-sum games?
- 6 Why is the computable concept the reached one?

- 7 How does this answer Module 03?
- 8 Why do the price of anarchy bounds survive?
- 9 When does best-response dynamics converge, and when does it cycle?
- 10 Why must a mechanism be tested against strategic agents?